Integrating Worldwide University Data from Wikidata, ROR and DEQAR

Project Report – Web Data Integration, HWS 2026

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Abstract. We integrate three open sources on universities: Wikidata, the Research Organization Registry (ROR) and the Database of External Quality Assurance Results (DEQAR). The 61,807 records describe 44,340 distinct institutions, 14,630 of them in at least two sources and 2,837 in all three. We translate the sources into one XML schema with 21 attributes, build a gold standard from the identifier links of the sources and from hand-picked corner cases, compare blocking strategies, similarity measures, rules and learned matchers with PyDI, and fuse the matched records with seven conflict resolution strategies. Learned matchers reach an F1 of 96–99% on the gold standard, 4–8 points above the best matcher that uses names only; the fused dataset fills attributes such as the founding year for 99% of the entities that are in at least two sources.

1 Introduction

Students, researchers and policy makers who compare universities worldwide meet registries, knowledge bases and quality assurance databases that differ in format, definitions and coverage. ROR has reliable names and locations but few descriptive facts, Wikidata adds founding years, student numbers and memberships but is incomplete, and DEQAR records mergers, closures and external quality assurance reports, mostly of European institutions. Our goal is one dataset that describes every university exactly once, with the best value for each attribute.

As in the project abstract, a *university* is a higher education institution that awards academic degrees and is an organisation of its own; a college or institute that meets both conditions counts, a branch, faculty or campus does not. The sources draw this line differently (ROR also holds branches, DEQAR faculties and closed predecessors), and none of them records whether an institution awards degrees. We therefore keep such records in the input and use them as hard non-matches of the gold standard (Sect. 3.1) instead of filtering them by rules the data cannot support. The integration follows the three phases of the course: data collection and translation (Sect. 2), identity resolution (Sect. 3) and data fusion (Sect. 4). All code, the translated data, the gold standards and the fused dataset are part of the submission (folders scripts/ and data/).

Table 1. Datasets

Dataset	Source (*)	Format	# entities	# attributes	List of attributes (**)
Wikidata	query.wikidata.org/sparql	CSV	22,697	16	Wikidata id, name, alt. names (MV), acronyms (MV), country, country code, city, website, founding year, latitude (MV), longitude (MV), students (MV), types, member of (MV), ROR id (MV), Wikipedia title (MV)
ROR	doi.org/10.5281/zenodo.22902037	CSV	30,131	18	ROR id, name, alt. names, acronyms (MV), country, country code, region, city, website, domains, founding year, latitude, longitude, types, status, parent organisation (MV), Wikidata id, Wikipedia title (MV)
DEQAR	eqar.eu/qa-results/download-data-sets	CSV	8,979	13	DEQAR id, name, alt. names (MV), country, city, website, founding year (MV), closure year (MV), parent institutions (MV), history (MV), QA reports (MV), last QA report (MV), ROR id (MV)

(*) Retrieved with our own scripts (scripts/fetch_*.py, build_ror_education.py). (**) (MV) = more than 30% missing values. The identifiers and links to other sources are used only as gold standard.

2 Phase I: Data Collection and Translation

Datasets. Our Python scripts retrieved Wikidata and ROR on 2 October 2026 and DEQAR on 9 October 2026 and store each source as one CSV file with one row per institution; attributes with several values are lists. *Wikidata* [Vrandečić and Krötzsch, 2014] comes from SPARQL queries against query.wikidata.org/sparql: all items that are an instance of *university* (Q3918) or one of its subclasses, with identifiers replaced by English labels. *ROR* [ROR, 2026] is the data dump v2.13 of 22 September 2026; of its 141,528 records we kept the 30,131 of type *education* that are not withdrawn. *DEQAR* [EQAR, 2026] is the list of higher education institutions that EQAR publishes as a nightly CSV file; we merged rows that repeat an institution once per location and took the country from the ETER identifier where the file has none. The three sources are edited by volunteers (Wikidata), a curation team (ROR) and quality assurance agencies (DEQAR). On linked records the founding year of DEQAR equals that of ROR in only 73% and that of Wikidata in 74% of the cases, so the sources are sufficiently distinct to vote against each other. Table 1 profiles them.

Sources considered but not used. DBpedia [Lehmann et al., 2015] is extracted from the English Wikipedia, on which Wikidata partly draws as well: 63% of its linked pairs with Wikidata have coordinates identical to about ten metres, so it is not distinct enough. *OpenStreetMap* [OpenStreetMap contributors, 2026] (47,706 universities) is independent but gives little beyond name and position (founding year for 2%). The *THE ranking* [THE, 2025] has only 3,118 institutions and shares only name, country and student numbers, and the *Hipo domain list* [Hipo, 2026] adds nothing that our sources lack.

Table 2. Attribute intersection with the integrated schema (W = Wikidata, R = ROR, D = DEQAR)

Attribute	Type	Found in	Attribute	Type	Found in
name	string	W, R, D	parent organisation	string/list	R, D
alternative names	string/list	W, R, D	region, status	string	R
country	string	W, R, D	internet domains	string/list	R, W, D
city	string	W, R, D	institution types	string/list	W
website	string	W, R, D	number of students	integer	W
founding year	integer	W, R, D	memberships	string/list	W
acronyms	string/list	W, R	closure year	integer	D
country code	string	W, R	history	string/list	D
latitude, longitude	decimal	W, R	QA reports, last date	integer, date	D

Integrated schema. We applied the rules of thumb for schema integration to the attribute lists of Table 1. The integrated schema has one class, *University*, with the 21 attributes of Table 2; 11 are in at least two datasets, five besides the name are in all three (alternative names, country, city, website, founding year) and seven are lists. Two mappings are not one-to-one: ROR names a parent organisation by identifier, which we replace by its name, and the *types* of ROR are registry categories such as *education* or *funder* and are therefore not mapped to the institution types of Wikidata. The internet domain is derived from the website for Wikidata and DEQAR. No attribute is a key; the source identifiers are not part of the schema, and a university is identified by name and country, which is unique for 99.1% (Wikidata), 99.8% (ROR) and 94.5% (DEQAR) of the records.

Schema mapping and translation. We first defined the target schema as an XSD (data/target.xsd): the elements *id, name, alternativeNames, acronyms, location, web, lifecycle, parentInstitutions, institutionTypes, numberOfStudents, memberships, qualityAssurance* and *status*. A first MapForce mapping for Wikidata and DEQAR had flaws: ROR was not translated, the lists were split with stray spaces, the institution types were not split, names and countries were lowercased, and the DEQAR parent institutions, which are needed for the hard nonmatches, were missing. So that all three sources follow exactly the same rules, the final files data/Wikidata_Mapping.xml, data/ROR_Mapping.xml and data/deqar_Mapping.xml are written by one script (translate_to_xml.py) with the element structure of the first mapping. They contain 22,697, 30,131 and 8,979 institutions, keep the original spelling of names and countries, and validate against the XSD. All following steps read these XML files.

3 Phase II: Identity Resolution

3.1 Gold standard

The sources link each other through identifier columns (Wikidata with ROR, DEQAR with ROR); two records are linked across Wikidata and DEQAR if they point to the same ROR record. We use the links that connect two records one-to-one as *matches*: 13,020 Wikidata–ROR, 3,391 DEQAR–ROR and 2,144

Table 3. Gold standard: matches and non-matches by kind

Pair	Matches	Hard	Random	Part of	Predecessor	Same name
Wikidata–ROR	13,020	13,020	6,510	2,154	0	229
DEQAR–ROR	3,391	3,391	1,695	969	270	90
Wikidata–DEQAR	2,144	2,144	1,072	237	190	80

Table 4. Blocking: pair completeness (PC) and number of candidate pairs

Strategy	Wikidata–ROR		DEQAR–ROR		Wikidata–DEQAR	
	PC (%)	Cand.	PC (%)	Cand.	PC (%)	Cand.
Standard: normalised name	75.7	11k	50.3	2k	59.9	2k
Standard: country + first word	83.8	872k	68.2	148k	75.7	101k
Sorted neighbourhood (window 10)	84.1	241k	68.0	101k	75.9	95k
Token: words of the name	–	237M	–	91M	–	84M
Token: country-scoped name tokens	99.2	597k	99.1	193k	98.6	110k
Token (scoped) + same web domain	99.7	597k	99.6	194k	99.5	110k

Wikidata–DEQAR pairs. The links are only used for the gold standard; the identifiers never enter the matching. They are incomplete, so we do not derive non-matches from missing links. Table 3 shows the *non-matches*: *hard* are candidate pairs of the most similar names in which both records are linked to a different record of the other dataset (so neither can be the other's partner), *random* a random draw of such pairs, and three kinds of corner cases that the proposal singled out: a *part of* an institution (a ROR record with a parent) against the institution or against another part, a closed DEQAR institution against the successor named in its history (*predecessor*), and *same name* in different countries. The pairs are split by their left record into a development part (tuning) and a test part (reported), so that no record is in both.

3.2 Blocking

Comparing all records would take 0.7 billion pairs for Wikidata–ROR. We compared the blockers of PyDI [Web-based Systems Group, University of Mannheim, 2026], which implement the standard techniques [Christen, 2012], by *pair completeness* (share of the linked pairs that survive blocking) and number of candidate pairs (Table 4). Equality of the normalised name finds only half to three quarters of the matches, because 25–52% of the linked pairs differ in name even after normalisation (local-language names, abbreviations); a country-and-first-word key and sorted neighbourhood reach 68–84%. Token blocking on the words of the names produces hundreds of millions of pairs because words such as *university* are everywhere. Our best strategy scopes the tokens of the name, alternative names and acronyms to the country (`DE_heidelberg`), drops tokens that occur in more than 150 records of a country, adds rare tokens without country for records whose country differs or is missing, and adds all pairs that share an internet domain. It keeps 99.5–99.7% of the matches with 0.11–0.60 million candidate pairs (reduction ratio above 99.9%).

Table 5. Matching on the test part of the gold standard (precision, recall, F1 in %); thresholds and weights tuned on the development part

Matcher	Wikidata–ROR			DEQAR–ROR			Wikidata–DEQAR		
	P	R	F1	P	R	F1	P	R	F1
Name, Jaro-Winkler	96.4	77.0	85.6	90.2	54.7	68.1	93.3	65.7	77.1
Name, Levenshtein	97.4	76.7	85.8	91.6	54.4	68.3	96.6	62.4	75.8
Name, Jaccard	94.5	77.8	85.3	95.2	55.0	69.7	96.0	66.7	78.7
Name, Sørensen-Dice	94.5	77.8	85.3	95.2	55.0	69.7	96.0	66.7	78.7
All names, best match, Jaro-Winkler	93.2	95.2	94.2	95.9	80.8	87.7	95.3	82.5	88.4
All names, best match, Jaccard	93.5	95.0	94.3	95.8	82.3	88.5	95.5	83.2	88.9
names	93.2	95.2	94.2	91.2	82.5	86.6	95.3	82.5	88.4
names+country	98.0	94.4	96.2	95.8	80.6	87.6	95.3	82.5	88.4
names+country+domain	95.9	98.1	97.0	97.7	93.3	95.4	96.8	94.7	95.7
names+country+domain+city	95.8	98.1	97.0	96.4	92.9	94.6	96.8	94.7	95.7
all attributes	96.5	93.0	94.7	98.3	90.1	94.1	96.2	94.7	95.4
logistic regression	97.9	97.3	97.6	97.7	93.6	95.6	97.4	94.9	96.1
random forest	99.1	98.2	98.6	98.9	93.6	96.2	96.6	96.1	96.4
gradient boosting	99.5	97.6	98.6	98.3	94.2	96.2	98.1	94.1	96.1

3.3 Matching

Comparators. For every pair we compute 11 similarities with PyDI's comparators: the name with four string measures (Jaro-Winkler, Levenshtein, Jaccard on words, Sørensen-Dice), the best match over all names of the two records (name, alternative names, acronyms) with Jaro-Winkler and with Jaccard, the country (1 if equal, 0.5 if unknown, 0 if different), the shared internet domain, the city (Jaro-Winkler), the founding year (absolute difference scaled to 10 years) and the distance between the coordinates (1 up to 2 km, 0 from 50 km). A missing value gives 0. Four indicator features tell whether both records have a domain, city, year or position.

Experiments. We compared (i) single similarity measures on the name, (ii) five rule-based matchers that are the weighted average of an increasing set of attributes (PyDI's `RuleBasedMatcher`), with integer weights from 0 to 6 found by random search on the development part, and (iii) three learned matchers (logistic regression, random forest, gradient boosting of scikit-learn [Pedregosa et al., 2011], in PyDI's `MLBasedMatcher`) on all 15 features. Thresholds were tuned on the development part (learned matchers: five-fold cross validation). Table 5 shows precision (P), recall (R) and F1 on the test part. As in PyDI's evaluator, a predicted pair is a false positive only if the gold standard labels it as a non-match; matches lost by blocking count as missed.

Findings. (1) The choice of the string measure hardly matters (F1 85.3–85.8% for Wikidata–ROR), but comparing *all* names of two records instead of the main names raises F1 by 8–19 points, because the main names often differ by language. (2) The web domain is the most useful further attribute (+0.8 to +7.8 points on top of names and country); the country helps most for Wikidata–ROR (+2.0 points), where names such as "Victoria University" exist in several countries. (3) Adding all attributes to a weighted average does *not* help ("all attributes" is 0.3–2.3 points below the rule with names, country and domain): the

founding year and the coordinates are missing for many records, and a missing value counts as 0 and lowers the score of true matches. (4) Learned matchers, which see the indicator features, are best by 0.7–1.6 points over the best rule: random forest for Wikidata–ROR and gradient boosting for DEQAR–ROR and Wikidata–DEQAR were chosen by their development F1. The domain is the most important feature of the DEQAR pairs (importance 0.36 and 0.80), then the best name match. Recall at the final threshold (0.8, see below) is 97%, 93% and 95% for test matches in which both records have a website but 86%, 64% and 47% when one has none (414, 25 and 19 matches). (5) The final threshold misses 3.5%, 7.4% and 5.6% of the test matches. They are mostly records with names in different languages and different websites (e.g. *Academy of fine arts of Roma – Accademia di Belle Arti di Roma*) or a campus against its university (*Oklahoma State University – Oklahoma State University-Main Campus*); 5–7% of the missed matches were already lost by blocking.

3.4 One-to-one matching and the final threshold

Every institution should be matched at most once per dataset, so we compared PyDI's post-clustering algorithms (Table 6). Greedy and stable one-to-one matching give the same result and raise precision to 99.1–100% at a recall loss of 0.6–1.0 points; maximum bipartite matching is worse. We use greedy matching.

The F1-optimal thresholds were found on a gold standard whose non-matches are mostly easy. We therefore checked, for each dataset pair, 60 randomly drawn correspondences that are *not* in the gold standard by hand (Table 7, columns "Sample"). At the F1-optimal threshold only 75% of the new Wikidata–ROR pairs were correct, the errors being similar institutions in one city (*British University of Nicosia – Cyprus Aydin University*), a part against the whole (*Kmct Polytechnic College – KMCT Group of Educational Institutions*) and predecessors (*University of Auvergne – Université Blaise Pascal*). A threshold of 0.8 raises the precision of the new pairs to 97% (Wikidata–ROR), 94% (DEQAR–ROR) and 98% (Wikidata–DEQAR) for a loss of at most 1.6 points of recall on the gold standard. We use 0.8 for all pairs. The resulting 20,079 correspondences contain 2,318 pairs that are not in the gold standard, i.e. links that the sources do not give; the sample suggests that nearly all of them are correct.

4 Phase III: Data Fusion

Groups. We merge the three correspondence sets into groups (connected components) with PyDI. Because the sets are matched pairwise, they can contradict each other (two Wikidata records linked to one ROR record through DEQAR); we therefore accept correspondences from the highest score down and reject one that would put two records of the same dataset into a group. This rejected 49 of 20,079 correspondences. The 61,807 records form 44,340 groups, i.e. **44,340 entities**: 15,771 only in ROR, 9,465 only in Wikidata, 4,474 only in DEQAR, 10,125 in ROR and Wikidata, 1,398 in DEQAR and ROR, 270 in DEQAR and Wikidata and 2,837 in all three.

Table 6. Post-clustering of the correspondences (threshold of the matcher, test part of the gold standard)

Pair	Post-clustering	Correspondences	P	R	F1
Wikidata–ROR	no post-clustering	14,625	99.1	98.2	98.6
	Greedy one-to-one	13,293	100.0	97.6	98.8
	Maximum bipartite matching	13,301	99.7	95.1	97.4
	Stable matching	13,290	100.0	97.6	98.8
DEQAR–ROR	no post-clustering	4,818	98.3	94.2	96.2
	Greedy one-to-one	4,278	99.6	93.6	96.5
	Maximum bipartite matching	4,281	99.1	90.7	94.7
	Stable matching	4,277	99.6	93.6	96.5
Wikidata–DEQAR	no post-clustering	3,666	98.1	94.1	96.1
	Greedy one-to-one	3,010	99.1	93.1	96.0
	Maximum bipartite matching	3,014	99.0	91.4	95.0
	Stable matching	3,008	99.1	93.1	96.0

Table 7. Threshold analysis after greedy one-to-one matching: gold standard (test part) and hand-checked sample of the correspondences that are not in the gold standard

Pair	Threshold	Corr.	P	R	F1	Sample	Sample precision
Wikidata–ROR	0.5	13,293	100.0	97.6	98.8	60	75.0
	0.7	13,048	100.0	96.8	98.4	42	92.9
	0.8	12,912	100.0	96.0	98.0	39	97.4
	0.9	12,609	100.0	93.7	96.8	32	100.0
DEQAR–ROR	0.5	4,278	99.6	93.6	96.5	60	88.3
	0.7	4,231	99.7	93.0	96.2	58	91.4
	0.8	4,157	99.7	92.0	95.7	54	94.4
	0.9	4,039	99.7	89.8	94.5	49	93.9
Wikidata–DEQAR	0.5	3,010	99.1	93.1	96.0	60	98.3
	0.7	3,010	99.1	93.1	96.0	60	98.3
	0.8	3,010	99.1	93.1	96.0	60	98.3
	0.9	2,929	99.3	92.5	95.8	57	98.2

Strategies. The conflict resolution functions follow the usual taxonomy [Bleiholder and Naumann, 2009]. For the list attributes (alternative names, acronyms, domains, parent institutions, types, memberships, history) we take the union. For the single-valued attributes we compared seven strategies with PyDI's fusion engine: PyDI's default (first non-null value), *favour a source* with each of three orders (ROR, Wikidata or DEQAR first; PyDI's `prefer_higher_trust`), and majority *voting* on normalised values (own resolver) with ties decided in favour of ROR, in favour of Wikidata, or of the earliest year. Coordinates are fused with the median of the values. Attributes of only one source need no resolution. In 99.4% of the groups with two country codes the sources agree, but only 70% agree on the name, 74% on the city, 80% on the founding year and 86% on the web domain; the rest are the conflicts that fusion has to resolve. With two sources most votes are ties, so a vote is in effect a trust-based choice among the disagreeing pairs.

Evaluation. We evaluated the strategies on a fusion gold standard of 39 well-known universities that are in all three datasets (data/gold_standard/fusion_gold.csv); the expected name (English or official local name), country, city, founding year and web domain were compiled by hand from general knowledge of these institutions, not from one of the sources. Table 8 shows the accuracy.

Table 8. Accuracy (%) of the fusion strategies on the 39 entities of the fusion gold standard

Strategy	Name	Country	City	Founding year	Domain
first non-null (PyDI default)	92.3	100.0	100.0	89.7	97.4
ROR first	100.0	100.0	100.0	94.9	97.4
Wikidata first	100.0	100.0	87.2	97.4	100.0
DEQAR first	92.3	100.0	100.0	89.7	97.4
vote, tie: ROR	100.0	100.0	100.0	94.9	100.0
vote, tie: Wikidata	100.0	100.0	100.0	94.9	100.0
vote, tie: earliest year	100.0	100.0	100.0	94.9	100.0

Table 9. Share of entities with a value (%): source datasets and fused dataset over all records, and best single source versus fused record for the entities that are in at least two sources

Attribute	Wikidata	ROR	DEQAR	Fused	Best source	Fused
	all entities				entities in ≥ 2 sources	
Alternative names	53	71	69	66	76	91
Acronyms	12	54	0	40	60	65
Country code	95	100	99	98	98	100
City	83	100	74	89	98	100
Region	0	100	0	68	98	98
Web domain	75	97	88	85	97	99
Founding year	81	89	59	80	94	99
Coordinates	69	100	0	79	98	99
Institution types	100	0	0	51	90	90
Number of students	16	0	0	8	23	23
Memberships	22	0	0	11	33	33
Parent institutions	0	9	28	12	7	12
QA reports	0	0	57	11	22	22

Voting is as good as or better than every rule that favours a source, except for the founding year. Favouring DEQAR, which is also what PyDI's default does here, is least accurate (founding year 89.7%, name 92.3%: three official local names are not among the accepted names); favouring Wikidata sometimes takes a district as the city (Wikidata gives "Taka-Töölö" for the Sibelius Academy in Helsinki) and is wrong for 13% of the entities. For the founding year, Wikidata first (97.4%) is one entity better than voting (94.9%), which is not a reliable difference on 39 entities. Our final strategy takes, per attribute, the best accuracy and decides ties in favour of voting: voting for name, country, city and domain, and Wikidata first for the founding year. The set is small and its universities are well documented, so it can show that a strategy is bad but not rank the good ones.

Quality of the fused dataset. Fusion fills the gaps of the sources (Table 9). For the entities that are in at least two sources the fused record has a founding year in 99%, a city in 100%, coordinates in 99% and alternative names in 91%, where the best single source has 94%, 98%, 98% and 76%. Over all 44,340 entities, attributes that only one source provides remain sparse (students 8%, memberships 11%, quality assurance reports 11%) because only few of their entities are in that source.

Table 10. Project requirements (slide 12) and how the project meets them

Requirement	Our project
3 different data sets	3: Wikidata, ROR, DEQAR 2.
44,340 entities (61,807 records) 3.	at least 1,000 entities in at least two datasets
at least 8 attributes; entities identifiable by at least two attributes	21; name + country 5.
11; 5 other than the name in all three 6.	ideally at least one list attribute

1. at least 2,500 entities in total (good: $> 10{,}000$ but $< 100{,}000$) 14,630 (2,837 in all three) 4. at least 5 attributes in at least two datasets; some in three 7, e.g. alternative names, history

5 Conclusion

All requirements of the course are met: three datasets from distinct sources, 44,340 entities (61,807 records), 14,630 entities in at least two and 2,837 in all three datasets, 21 attributes of which five besides the name are in all three, and seven list attributes. The main lessons are: names alone are not enough across languages (the best match over all names and the web domain matter most); learned matchers beat hand-weighted rules mainly because they handle missing values; a gold standard whose non-matches are easy gives thresholds that are too low, which a hand-checked sample of the correspondences revealed; and with only two sources voting is mostly a trust decision.

Limitations. The matches of the gold standard are the pairs that the sources link themselves, so they are over-represented among records with a website and

a Wikidata or ROR identifier, and the recall on unlinked records is unknown. The hand-checked samples (60 pairs per dataset pair) give a precision with a margin of roughly ±6 points, and the fusion gold standard has 39 entities. Parts and predecessors are kept as separate entities but the data cannot tell a branch from a university in general, so non-universities of ROR and DEQAR remain in the fused dataset.

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